

Robot Electronics Chat Summary

Date: 2026-05-18

Project: Dome2 hardware / robotics electronics documentation

Primary file: `robot_electrical_components.md`

Session Purpose

This session focused on creating and tightening a concise electrical and electronic component document for the robot. The document is intended to support compatibility analysis, power budgeting, wiring decisions, signal-level checks, and eventual Linorobot2/ROS integration.

Work Completed

Created and refined `robot_electrical_components.md` as a compact engineering reference. The document now tracks:

- Raspberry Pi 4 Model B
- Raspberry Pi I2C shim
- Waveshare 3S UPS module
- Panasonic NCR18650GA 18650 cells
- System power architecture
- Compatibility risks and open design questions

The document was tightened to remove repeated structure and keep the important engineering concerns visible.

Current Document Structure

`robot_electrical_components.md` now uses these sections:

- Purpose
- Components
- Component Inventory
- Power Architecture
- Compatibility Notes
- Open Questions

Key Technical Notes Captured

- Raspberry Pi 4B is powered from the regulated 5 V output of the Waveshare UPS module.
- The Waveshare UPS module accepts a 3S battery pack and provides up to 5 A / 25 W on the 5 V rail.
- Motor power should come directly from the 3S battery voltage, not from the UPS 5 V rail.
- All control electronics should share a common ground.
- GPIO and I2C connections to the Raspberry Pi must be 3.3 V safe.
- I2C address conflicts need to be checked once devices are selected.

- A 3S Panasonic NCR18650GA pack is estimated at about 38 Wh.

Runtime Estimates Captured

Average Power	Estimated Runtime
10 W	~3.8 h
20 W	~1.9 h
30 W	~1.3 h

Open Questions

The current open questions are:

- Exact Raspberry Pi RAM size?
- Which USB devices will be attached?
- Which GPIO pins are reserved?
- Exact I2C shim model?
- Which I2C devices and addresses will be used?
- Which charger/power adapter will be used?
- Which peripherals will draw from the 5 V rail?
- Will battery monitoring be integrated into ROS?
- Are the three installed cells new and matched?
- What is the expected average system power draw?

Transcript Export Guidance Discussed

Because the ChatGPT UI may not expose a **Copy conversation** command in all versions, the practical export options discussed were:

- Select the chat manually with **Cmd+A**, copy with **Cmd+C**, and paste into a Markdown file.
- Use the Share feature, then save the shared page contents.
- Use **Cmd+P** and save the chat as PDF.
- Use ChatGPT Settings -> Data Controls -> Export Data for a full account export.

Recommended Repository Organization

The suggested documentation layout was:

```
docs/  
  robot_electrical_components.md  
  robot_chassis_design.md  
  chat_transcripts/
```

In this workspace, the session summary was saved as:

docs/chat_transcripts/2026-05-18-robot-electronics-summary.md

Follow-Up Work

Useful next steps:

- Fill in the unknown Raspberry Pi model details and attached USB devices.
- Identify exact motor driver and motor current requirements.
- Add a formal 5 V power budget table.
- Add a direct battery-voltage power budget for motors.
- Confirm UPS module model and battery monitoring interface.
- Confirm whether the 18650 cells are matched and appropriate for a 3S pack.